

Rajesh Dawadi

rajesh.dawadi@gmail.com, 700 Laguna St, Apt 110, San Francisco, CA, 94102

Work & Research Experience

Over 10 years of experience in software development and research expertise in areas including communication network protocol development, augmented reality (AR) software

Collaborations in 5 published research articles at reputed IEEE forums and proposed 2 patents

Software Developer 10/23/2023 - Present
Meta Platforms, Burlingame, CA, USA

- Worked on Applied AI Engineering to improve Meta-internal model
- Experience with features development related to real-time communication applications for voice and video calling (Augmented Reality Communication platform)
- Troubleshooting and debugging issues seen on various devices on the communication stack
- Programming experience in C++, Kotlin, Javascript

Software Developer 02/10/2020 - 10/20/2023
Cisco Systems, Kanata, Ottawa, Canada

- Network software developer working on Cisco Data Center family of switch products (N7000, N9000) based on the Nexus Operating System
- Experience with leading features development and troubleshooting related to networking protocols as part of the BGP development team.
- Knowledge of various network virtualization overlay technologies, such as VxLAN, MP-BGP EVPN, Segment Routing v6 (SRv6), etc
- Programming experience in C/C++, Python

Radio Software Developer 10/02/2017 - 02/07/2020
Ericsson, Kanata, Ottawa, Canada

- Worked on embedded radio software development for Ericsson 4G/5G wireless products
- Worked on Linux Device Driver for power management module
- Familiarity with W-CDMA, LTE, ORAN, worked on Self-Test framework
- Debugged and fixed customer issues relating to various radio products

Embedded Software Designer 01/17/2017 - 09/29/2017
Evertz Microsystems, Burlington, Canada

- Worked on device driver development in C/C++ for 10G SFP, video/audio broadcasting and networking products
- Developed and maintained applications for adding/monitoring ethernet, SFP network bridge on video hub

- Porting various features (DHCP, Syslog, LLDP) on audio/video routers.
- Experience using stack trace and memory analyzer tools like gdb, valgrind

LTE Systems Engineer

06/25/2012 - 08/04/2014

Reliance Jio Infocomm Limited, Mumbai, India

- Worked as a system engineer on LTE specifications, performed TD-LTE eNodeB protocol level validation & advanced features testing
- Conducted Rf Planning on the LTE network
- Performed Voice over LTE, Self-optimizing networks (SON), Load balancing testing

Research Assistant

09/02/2014 - 08/29/2016

Broadband Wireless Communications Lab, McGill University

- Worked as a research assistant at the Broadband Wireless Communications Lab at McGill University during Master's Thesis study
- Research areas: Resource allocation in virtualized wireless networks, optimization, dynamic user association in next generation wireless networks.

Expertise and skills

Programming: C, C++, Python, Matlab, Bash Scripting, Linux.

Version Control: Git, SVN.

Projects:

- Developed a sliding window based Load Balancer using eBPF hook on the kernel and deployed on containers.
- Network simulator based on C++ : Implemented a multi-threaded link layer network simulator on C++
- Simple DNS spoof daemon to intercept and send spoof DNS reply based on raw packet capture from Linux network implementation in C
- Weather notification simulator: Developed a client-server type notification system on C++
- RSS feed reader: Developed a news feed reader in Python that could catch news items related to certain configurable filters and display the latest ones

EDUCATION

Master of Engineering, Electrical Engineering,

Dept of Electrical & Computer Eng, McGill University, Canada

Date of Completion: 08/30/2016, Degree awarded: May, 2017

- Thesis: User-Association and Resource Allocation in Multi-cell Virtualized Wireless Networks
- Research Area: Resource allocation in Wireless Networks, Virtualization in Wireless Networks

Bachelor of Technology, Electronics and Communication Engineering

National Institute of Technology, Bhopal, India

Aug 2008 - Apr 2012, Degree awarded: Oct, 2012

- Courses: Computer networks, Computer Architecture, C programming, Mobile Communications, Digital Electronics, Digital Signal Processing.

Patents proposed

- "Industrial Automation with Cellular Network", E. Iun, E. Viera, R. Dawadi, 2019 (Patent: patentimages.storage.googleapis.com/f9/07/0a/59b368bbe21dca/US20220294881A1.pdf)
- "Software Defined Manufacturing in a cellular network", E. Vieira, E. Iun, R. Dawadi, T. Tian, L. Wang, 2021 (<https://patentscope.wipo.int/search/en/detail.jsf?docId=WO2021024113>)

Publications

- "Joint user association and resource allocation in multi-cell VWN", S. Parseefard, R. Dawadi, M. Derakhshani, T. Le-Ngoc, IEEE Access, 2016: 102 citations (<https://ieeexplore.ieee.org/document/7462252>)
- "Dynamic Resource Allocation for Virtualized Wireless Networks in Massive-MIMO-aided and Front-haul-Limited C-RAN", IEEE Transactions on Vehicular Technology: 55 citations: S. Parseefard, R. Dawadi, M. Derakhshani, T. Le-Ngoc, M. Baghani (<https://ieeexplore.ieee.org/document/7940003>)
- "Energy-Efficient Resource Allocation in Multi-cell Virtualized Wireless Networks", R. Dawadi, S. Parseefard, M. Derakhshani, T. Le-Ngoc, IEEE ICUWB, Oct. 2015, 9 citations (<https://ieeexplore.ieee.org/document/7324446>)
- "Adaptive Pilot-Duration and Resource Allocation in Virtualized Wireless Networks with Massive MIMO", R. Dawadi, S. Parseefard, M. Derakhshani, T. Le-Ngoc, IEEE WCNC, Apr. 2016 (<https://ieeexplore.ieee.org/document/7564902>)
- "Power-efficient Resource Allocation in Non-Orthogonal Multiple Access (NOMA) virtualized wireless networks, 2016 IEEE Global Communications Conference (GLOBECOM): 34 citations, R. Dawadi, S. Parseefard, M. Derakhshani, T. Le-Ngoc, IEEE WCNC (<https://ieeexplore.ieee.org/document/7842162>)